

LME Metals Risk-Premia Strategies

Momentum, Carry, Value and the Combined Portfolio - Copper and Aluminium

Methodology, Research Process, and Performance

Backtest window: 2006-01-03 to 2025-12-31

1. Executive Summary

This document defines the four strategies presented on the dashboard - Momentum, Carry, Value, and the combined Portfolio - for both LME Copper and LME Aluminium. For each strategy it states the economic rationale, the exact signal formula, the trading assumptions, the research process used to select the winning configuration, and the realised performance over the full sample and over the trailing five years.

The central result is the diversification thesis at the heart of the framework: three economically distinct, near-uncorrelated return sources combine into a portfolio whose risk-adjusted return materially exceeds any single sleeve. For Copper the equal-weight portfolio earns a net Sharpe of 1.00 over the full sample (best single sleeve 0.70); for Aluminium it earns 0.85 (best single sleeve 0.64). The portfolio also cuts drawdown roughly in half versus the individual signals.

A second result is that the optimal configuration is metal-specific. Copper rewards a medium-speed trend and a fast curve-momentum carry; Aluminium rewards a slower trend and a mean-reverting (z-score) carry. The dashboard auto-switches to each metal's best configuration when the metal toggle is changed, and every number in this document is reproduced live there.

Headline figures (net of 5 bps round-trip transaction costs, active-day Sharpe):

Metal	Best single sleeve	EW Portfolio (full)	EW Portfolio (trailing 5y)	Walk-forward OOS
Copper	Momentum 0.70	1.00	0.92	0.68
Aluminium	Carry 0.64	0.85	0.61	1.12

2. Common Methodology and Assumptions

All four strategies share one backtesting engine. The conventions below apply uniformly so that sleeves are directly comparable and can be combined without distortion.

2.1 Data and the continuous front-month series

- **Source.** LME daily settlement prices, 2006 to present. The futures curve provides contracts F1 through F27 (Price, Volume, Open Interest); the cash and 3-month forward file provides the Cash and 3M series used by some carry variants.
- **Continuous price.** Trading and PnL use a single back-adjusted front-month series, F1_continuous, built by stitching successive front contracts with a ratio (multiplicative) back-adjustment across each roll. This removes the artificial price jump on roll dates so that daily returns are clean and continuous.

- **Cross-metal build.** The Aluminium F1_continuous is built with the identical four-phase roll logic as Copper; its return series correlates 0.9994 with the validated Copper build, confirming the pipeline transfers cleanly across metals.
- **Signal vs PnL.** Signals are computed on raw, observable prices (F1_raw or the relevant Fk). PnL is always earned on F1_continuous. Transaction costs are charged on F1_raw, the actual traded price.

2.2 Transaction costs

Costs are modelled as a round-trip charge on each change in position: $\text{cost} = |\Delta \text{position}| \times (\text{bps} / 10000 / 2) \times \text{F1_raw}$. The dashboard exposes 0, 5, 10 and 20 bps. Headline numbers in this document are net of 5 bps, with gross shown alongside. Low-turnover signals (z-score carry, slow trend) are barely affected; high-turnover signals are penalised, which is intentional.

2.3 Performance metrics

- **Active-day Sharpe.** annualised mean / standard deviation of daily returns $\times \sqrt{252}$, computed over active days only (days the strategy holds a non-zero position) so that flat periods do not dilute the risk-adjusted measure.
- **Annualised return.** mean daily PnL return $\times 252$.
- **Max drawdown.** the largest peak-to-trough decline of the cumulative return path (in cumulative return points).
- **% Flat.** share of days with no position, indicating how selective a signal is.

2.4 In-sample versus walk-forward (how robustness is tested)

Two lenses are reported throughout. The in-sample (IS) backtest uses the full history and is useful for ranking variants. The walk-forward out-of-sample (OOS) test re-applies a fixed configuration on a rolling basis - train on 1260 trading days (about five years), trade the next 252 days, roll forward - and reports the average out-of-sample Sharpe across those untouched one-year windows (the same convention shown on every dashboard tab, including the Portfolio OOS section). A configuration is only trusted when its OOS Sharpe holds up; signals that look strong IS but collapse OOS (the Value V2 Baz-Granger reversal is the textbook example) are flagged and kept off the default path.

Three methodological categories apply across the strategies, and the dashboard labels each walk-forward section accordingly:

- **Rolling walk-forward (true OOS).** Combination weights are re-estimated by max-Sharpe quadratic programming on each preceding five-year in-sample window, then applied blind to the following twelve months. Each out-of-sample year is genuinely held out from the estimation step. Only the Anchors plus IS-Optimised Weights variant in the Momentum tab qualifies here.
- **Parameter-free (unconditionally OOS).** Signal construction is fixed on structural or published grounds with no free parameters estimated from historical data. Applies to: the Baz-Granger CTA paper variant (S, L pairs from the published paper), the Anchors equal-weight blend (three structural speeds at equal weight), and the V1 roll-yield level signal (sign of the front spread, no fitting). All reported figures for these signals are unconditionally out-of-sample.
- **Fixed-parameter (sub-period decomposition).** Signal parameters (MA windows, contract, lookahead, carry horizon) were identified across the full 2006-2025 history. The walk-forward OOS panel shows annual performance under those fixed parameters over the post-burn-in period (from approximately 2011): a sub-period performance decomposition, not an independent held-

out test. Applies to: MA crossover, carry-momentum 20d, z-score 252d, V1 MA-reversion, and the combined portfolio. The IS Sharpe and the walk-forward figure differ primarily because the first five years (2006-2010) are excluded from the walk-forward period, and the averaging convention (per-window mean versus pooled) adds a small secondary effect.

3. Momentum Strategy

3.1 Economic rationale

Commodity futures trend: supply and demand shocks resolve slowly, so price moves persist over weeks to months. Momentum monetises this persistence by holding the metal in the direction of its recent trend. Because the trend reverses gradually, momentum has no same-day urgency.

3.2 Definition and formula

The production signal is a dual moving-average crossover on the raw front price:

$$\text{signal}(t) = \text{sign}[\text{MA_fast}(F1_raw, m) - \text{MA_slow}(F1_raw, n)]$$

position(t) = +1 when the fast average is above the slow (uptrend), -1 when below. The position is shifted one day for execution (Same-Day convention). PnL = position $\times$ $\Delta F1_continuous$.

Alternative families evaluated and shown on the tab: a single-scale CTA signal, the three-scale Baz-Granger CTA response (detailed below), and an equal-weight blend of three structural anchors MA(10,25)+MA(35,43)+MA(63,100).

3.2.1 Baz-Granger CTA response function (alternative trend signal)

This is the trend signal from Baz, Granger et al. (Man-AHL, "Dissecting Investment Strategies in the Cross Section and Time Series"). It is built per timescale and then blended. For each of three fast/slow exponential-moving-average pairs (S, L) = (8, 24), (16, 48), (32, 96) days:

Step 1 - raw trend: $x = \text{EWMA}_S(\text{price}) - \text{EWMA}_L(\text{price})$

Step 2 - price-vol normalise: $y = x / \text{StdDev}_{63}(\text{price})$

Step 3 - standardise: $z = y / \text{StdDev}_{252}(y)$

Step 4 - response function: $u = z \cdot \exp(-z^2 / 4) / 0.89$

- **x (raw trend).** the difference of a fast and a slow exponentially-weighted moving average of price; positive in an uptrend.
- **y (vol-normalised).** dividing by the 63-day rolling standard deviation of price puts the signal in volatility units, so it is comparable across price levels and regimes.
- **z (standardised).** dividing by its own 252-day (one-year) rolling standard deviation rescales the signal to roughly unit variance, so z is a standardised trend-strength score.
- **u (response).** the response u rises with z for weak-to-moderate trends, peaks near $z \approx \sqrt{2}$, then DECAYS for large $|z|$. Economically it cuts conviction when a trend is overextended (and so prone to reversal). The constant 0.89 normalises the response to roughly unit amplitude. The three per-timescale u values are averaged and the position is sign(average u).

3.2.2 Anchors equal-weight blend (Anchors EW)

Rather than choosing a single optimal MA pair, a natural robustness strategy is to average across several structurally-motivated speeds. The three anchor windows cover short, medium, and long horizons:

$position_EW(t) = mean[sign(MA(F1,10) - MA(F1,25)), sign(MA(F1,35) - MA(F1,43)), sign(MA(F1,63) - MA(F1,100))]$

- **MA(10,25).** MA(10,25): short-horizon, about 2-5 weeks. Captures rapid moves; higher turnover.
- **MA(35,43).** MA(35,43): medium-horizon, the empirically optimal single pair for Copper. The central anchor.
- **MA(63,100).** MA(63,100): long-horizon, about 3-5 months. Low turnover, slower to flip.

Each anchor produces a ± 1 binary position. The equal-weight average falls in $[-1, -2/3, -1/3, 0, +1/3, +2/3, +1]$ - seven discrete levels - so the combined position is continuous in steps of $1/3$ rather than binary. A position of $+1/3$ means two of three anchors disagree.

Dashboard also shows an IS-Optimised Weights variant: on each OOS window, quadratic-programming (max-Sharpe, $w \geq 0$, $\sum = 1$) is applied to the prior 5yr IS data to find the optimal blend of the same three anchors. The optimiser typically concentrates weight on MA(35,43) for Copper (the clear IS winner), so the EW and IS-opt variants are close in practice. Both are shown on the walk-forward OOS panel for transparency. The Anchors EW is NOT used in the combined portfolio; the single best MA (MA(35,43) Copper, MA(60,115) Aluminium) is used for the momentum sleeve.

3.3 Research process and selection

- **Parameter scan.** the dashboard scans an explicit set of known-good pairs plus a coarse grid of (fast, slow) windows, scoring each on full-sample gross Sharpe, then re-checks the leader under both timings and net of costs.
- **Metal-specific optimum.** Copper rewards a medium-speed trend - MA(35,43) is the clear winner. Aluminium's fast trend is weak; it wants a much slower filter, MA(60,115), reflecting a smoother, more contango-driven price process.
- **Robustness.** the MA(35,43) Copper edge survives the Same-Day to Lag-1 step and holds in walk-forward (OOS 0.47), so it is genuine trend, not noise.

3.4 Assumptions

- position is ± 1 (always in market, long or short); turnover is low so costs are minimal (net $\approx$ gross).
- Same-Day execution (signal close t , return t to $t+1$).
- no volatility targeting - sizing is handled at the portfolio layer.

3.5 Performance

Copper: MA(35,43). Aluminium: MA(60,115). Net figures are after 5 bps.

Metal / Period	Gross Sharpe	Net Sharpe	Ann. Return	Max DD	% Flat
Copper - full sample	0.72	0.70	+18.3%	-64%	1%
Copper - trailing 5y	0.62	0.60	+13.0%	-28%	0%
Aluminium - full sample	0.48	0.48	+10.7%	-50%	2%
Aluminium - trailing 5y	0.37	0.36	+8.7%	-50%	0%

Walk-forward OOS Sharpe (net 5 bps, average across the rolling 1yr windows): Copper +0.47, Aluminium +0.34.

Trailing-five-year behaviour.

Copper momentum remained the steadiest sleeve through 2021-2025, riding the post-COVID rally and the 2022 macro swings with a far shallower drawdown (-28%) than over the full history.

Aluminium's slow trend gave back some edge in the choppier recent regime (Sharpe 0.37 vs 0.48 full-sample) but stayed positive and fully invested, confirming the slower filter is the right choice for that metal.

4. Carry Strategy

4.1 Economic rationale

The shape of the futures curve carries information about physical tightness. Backwardation (front above deferred) signals scarcity and a positive expected roll return to a long; contango signals surplus. Carry trades the curve slope. Unlike momentum, the carry level is a state variable, so the timing question is genuinely strategy-specific.

4.2 Definitions and formulas

The base roll yield is the front-end slope:

$$\text{roll_yield}(t) = (F_1 - F_2) / F_1$$

Several signal constructions are evaluated on the tab:

- **V1 level.** position = sign(roll_yield). Honest Sharpe $\approx$ 0.10 on Copper - kept as the baseline.
- **V4 carry-momentum (20-day).** position = sign(roll_yield - roll_yield[t-20]). Trades the 20-day change in the slope; a steepening curve (backwardation building) flags physical tightening. Best walk-forward carry on Copper.
- **V3 z-score (252-day).** position = sign((roll_yield - mean_252) / std_252). Standardises the slope over a rolling year; clean, low-turnover, and the best carry on Aluminium.
- **V2 long-dated slope.** deferred-tenor slope $(F_j - F_k)/F_k$ across ten tenor pairs (F3-F15 ... F12-F24) - shown for completeness but not selected.

Carry uses Same-Day execution: the slope is a level, and the flip day itself tends to coincide with the large directional move, so waiting an extra day forfeits the signal.

4.3 Research process and selection

- **Copper.** on Copper the carry-momentum 20-day signal is the standout - net 0.47 IS and the best walk-forward of any carry variant - because it is additive to price momentum (position correlation only +0.12) rather than a second trend signal.
- **Aluminium.** the 20-day carry-momentum is effectively dead on Aluminium; the 252-day z-score is the winner there (net 0.64 IS, OOS 0.83), matching its smoother curve dynamics.

4.4 Assumptions

- position ± 1 ; PnL on F1_continuous; costs on F1_raw.
- Same-Day execution.
- the z-score needs a 252-day warm-up; carry-momentum needs 20 days.

4.5 Performance

Copper: carry-momentum 20-day. Aluminium: z-score 252-day. Net after 5 bps.

Metal / Period	Gross Sharpe	Net Sharpe	Ann. Return	Max DD	% Flat
Copper - full sample	0.52	0.47	+13.5%	-47%	0%
Copper - trailing 5y	0.71	0.64	+14.8%	-35%	0%
Aluminium - full sample	0.66	0.64	+14.1%	-50%	5%
Aluminium - trailing 5y	0.33	0.30	+7.9%	-50%	0%

Walk-forward OOS Sharpe (net 5 bps, average across the rolling 1yr windows): Copper +0.25, Aluminium +0.89.

Trailing-five-year behaviour.

Copper carry was the best-performing sleeve of the last five years (net 0.64), as repeated inventory squeezes and backwardation episodes in 2021-2024 played directly to the curve-momentum signal. Aluminium carry, very strong over the full sample, was more subdued recently (net 0.30) as the curve spent long stretches in stable contango; the z-score still added positive, low-turnover return and diversified the trend sleeve.

5. Value Strategy

5.1 Economic rationale

Value is the contrarian counterweight to momentum: when a price has stretched far from its own long-run anchor it tends to mean-revert. Value is deliberately the weakest stand-alone sleeve, but its negative correlation to momentum (about -0.2 to -0.4) makes it valuable inside the portfolio - it cushions exactly when trend reverses.

5.2 Definitions and formulas

- **V1 MA-reversion.** deviation = $(F_k - MA_N) / MA_N$ on a deferred contract F_k . Position is +1 (buy) when the deviation falls below -10%, -1 (sell) when it rises above +10%, and flat in the $\pm 10\%$ band. This is the production signal.
- **V2 Baz-Granger reversal.** the long-horizon reversal signal from Baz, Granger et al. Detailed below. Higher in-sample Sharpe at long lookbacks but collapses out-of-sample - retained on the tab as a cautionary alternative, never the default.

Value uses Lag-1 / Same-Day with no urgency; the signal evolves slowly. The $\pm 10\%$ deadband keeps the V1 strategy flat (and cost-free) for long stretches, trading only genuine dislocations.

5.2.1 Baz-Granger long-horizon reversal (V2, formula detail)

Where momentum exploits short-horizon trend persistence, the Baz-Granger "value" leg exploits the opposite effect at long horizons: prices that have run a long way tend to mean-revert. The construction is deliberately simple - the signal is the price change over a long lookback N , sign-reversed so it is contrarian:

$$reversal(t) = F1_raw[t - N] - F1_raw[t]$$

$$position(t) = \text{sign}(reversal(t)) = +1 \text{ if the price has FALLEN over the last } N \text{ days, } -1 \text{ if it has RISEN.}$$

- **Lookback N.** N is the lookback in trading days, tested at 1, 3, 5, 7 and 10 years (252, 756, 1260, 1764, 2520 days). Longer N targets slower, more structural reversals.

- **Sign logic.** reversal > 0 means today's price is below where it was N days ago - the contract is judged "cheap" and the rule goes long (+1); reversal < 0 means it has appreciated and is judged "rich", so the rule goes short (-1). It is always in the market (no deadband, unlike V1).
- **Execution.** the position is shifted one day for execution (no look-ahead); PnL = position × $\Delta F1_{\text{continuous}}$, with transaction costs on $F1_{\text{raw}}$ on each flip.

The non-monotonic behaviour by lookback (3yr and 10yr work, 5yr is a trap with negative Sharpe) and the out-of-sample collapse are exactly why V2 is shown only as a cautionary alternative and V1 MA-reversion is the production value signal.

5.3 Research process and selection

- **Reference point.** the NGL energy risk-premia paper uses F12 with a five-year anchor. That contract is optimal for energy but not for Copper.
- **Metal-specific optimum.** sweeping the contract (F1, F5, F8, F12) and lookback (1, 3, 5, 7, 10 years), Copper's empirical optimum is F8 at a five-year anchor; Aluminium's is F12 at five years (closer to the energy convention).
- **IS vs OOS.** the V1 MA-reversion is robust out-of-sample (Copper OOS 0.36), whereas the V2 reversal's in-sample strength is a 2020-21 dislocation mirage that disappears OOS. V1 is therefore the default; V2 is shown only to demonstrate why it is not used.

5.4 Assumptions

- three-state position (+1 / 0 / -1) with a $\pm 10\%$ deadband; large flat share by design.
- five-year (1260-day) moving-average anchor, minimum 630-day warm-up.
- PnL on $F1_{\text{continuous}}$; costs on $F1_{\text{raw}}$; low turnover.

5.5 Performance

Copper: V1 F8, 5-year anchor. Aluminium: V1 F12, 5-year anchor. Net after 5 bps.

Metal / Period	Gross Sharpe	Net Sharpe	Ann. Return	Max DD	% Flat
Copper - full sample	0.28	0.27	+3.8%	-67%	46%
Copper - trailing 5y	-0.02	-0.03	-0.3%	-46%	25%
Aluminium - full sample	0.27	0.26	+3.1%	-68%	51%
Aluminium - trailing 5y	0.16	0.15	+2.4%	-55%	40%

Walk-forward OOS Sharpe (net 5 bps, average across the rolling 1yr windows): Copper +0.75, Aluminium +0.62. The per-window average flatters value because it trades selectively (flat ~40-50% of days); its full-sample (+0.27) and trailing-5y Sharpes are the more representative stand-alone figures.

Trailing-five-year behaviour.

Value is regime-conditional: most of its full-sample edge came from the 2020-2021 dislocation. Over the trailing five years Copper value was essentially flat-to-slightly-negative - a reminder it is not a stand-alone strategy - while Aluminium value stayed modestly positive (net 0.15). Its role is confirmed not by stand-alone Sharpe but by its diversification: it is the sleeve that turns contrarian when momentum is most exposed, which is exactly why it earns its third of the portfolio.

6. Combined Portfolio

6.1 Construction - Equal-Weight (default)

The three sleeves are combined daily into one position with a fixed one-third weight each:

$$Port_pos(t) = (1/3) \cdot Momentum(t) + (1/3) \cdot Carry(t) + (1/3) \cdot Value(t)$$

Each leg is the selected metal's best configuration, all on Same-Day execution. The combined position ranges from -1 (all three short) to +1 (all three long); a value such as +0.33 means two sleeves long, one short. The weights are fixed - they do not move with market conditions - which keeps the portfolio simple and transparent.

6.2 Construction - Inverse-Volatility (alternative)

The dashboard also offers an inverse-volatility (risk-balanced) weighting. Equal-weight equalises position SIZE, not risk: an always-on sleeve (momentum) contributes more variance than one that is flat 40% of the time (value). Inverse-vol corrects this by giving each sleeve a weight inversely proportional to its own recent volatility, so each contributes roughly equal risk.

$$\sigma_i(t) = \text{trailing 63-day std. dev. of sleeve } i\text{'s daily return, lagged one day}$$

$$w_i(t) = (1 / \sigma_i(t)) / [1/\sigma_m(t) + 1/\sigma_c(t) + 1/\sigma_v(t)]$$

$$Port_pos(t) = w_m(t) \cdot Momentum(t) + w_c(t) \cdot Carry(t) + w_v(t) \cdot Value(t)$$

- **Which volatility.** each sleeve's OWN realised daily strategy return (position $\times$ $\Delta F1_{\text{continuous}}$ / prior price), not raw price volatility. This measures how volatile that sleeve's P&L actually is.
- **Window.** a 63-trading-day (about three-month) trailing rolling standard deviation, lagged one day so today's weight uses only information available through yesterday - no look-ahead. (The 63-day window was itself chosen by a sweep: it is the Sharpe-optimal inverse-vol window; 10 days is too noisy, 126+ days gives up return for no Sharpe gain.)
- **Fixed or floating.** FLOATING, not fixed. The three weights are recomputed and renormalised to sum to 1 on every trading day, so the portfolio is rebalanced daily and tilts away from whichever sleeve is currently most volatile. During the first 63 days (volatility undefined) the weights fall back to 1/3 each.
- **Why it ends up near equal-weight.** because the three sleeves have similar realised volatility, $1/\sigma$ is similar for all three, so the average weights land near 1/3 each (the dashboard shows the live averages, e.g. about 0.34 / 0.34 / 0.39 for copper). Inverse-vol therefore behaves close to equal-weight: it earns a slightly higher annual return and is steadier post-2022, but equal-weight has a marginally better net Sharpe and a shallower drawdown, so equal-weight remains the default.

Every card, table and chart on the Portfolio tab recomputes for whichever weighting is selected; the figures in Section 6.4 below are the equal-weight default.

Legs by metal:

Sleeve	Copper	Aluminium
Momentum	MA(35,43)	MA(60,115)
Carry	Carry-momentum 20-day	Z-score 252-day
Value	V1 F8, 5-year anchor	V1 F12, 5-year anchor

6.3 Why it works - low correlation

The diversification claim rests on the three position series being close to uncorrelated. Measured over the full sample:

Pair	Copper	Aluminium
Momentum - Carry	+0.07	+0.23
Momentum - Value	-0.21	-0.36
Carry - Value	-0.05	-0.01

All pairwise correlations are below 0.25 in magnitude, and momentum-value is meaningfully negative. Combining three roughly independent sources of similar Sharpe lifts the portfolio Sharpe well above any single sleeve and roughly halves the drawdown.

6.4 Performance

Equal-weight portfolio, net after 5 bps.

Metal / Period	Gross Sharpe	Net Sharpe	Ann. Return	Max DD	% Flat
Copper - full sample	1.04	1.00	+11.9%	-27%	21%
Copper - trailing 5y	0.99	0.92	+9.2%	-9%	10%
Aluminium - full sample	0.87	0.85	+9.3%	-21%	20%
Aluminium - trailing 5y	0.64	0.61	+6.3%	-15%	20%

Walk-forward OOS Sharpe (net 5 bps, average across the rolling 1yr windows): Copper +0.68, Aluminium +1.12.

The portfolio dominates every individual sleeve on both metals and on both the full sample and the trailing five years. Its drawdown is the smallest of any line in this document (Copper -9% over the trailing five years versus -28% to -46% for the individual sleeves). The Aluminium walk-forward Sharpe of 1.12 - higher than its in-sample 0.85 - is strong evidence the combination is not over-fit.

6.5 Current positioning (latest available date)

Metal	Momentum	Carry	Value	EW Portfolio
Copper	+1 (long)	-1 (short)	-1 (short)	-0.33 (net short)
Aluminium	+1 (long)	+1 (long)	-1 (short)	+0.33 (net long)

Copper is net short (carry and value both bearish, trend still long); Aluminium is net long (trend and carry both bullish, value contrarian short). The split positioning across the two metals illustrates the framework operating independently per metal.

7. Cross-Metal Conclusions

- **The diversification thesis holds.** three economically distinct premia (trend, curve, mean-reversion) combine into a portfolio with materially higher Sharpe and roughly half the drawdown of any single sleeve, on both metals. This is the core of the framework.

- **Optima are metal-specific.** the same engine and the same three premia work on Aluminium, but the winning parameters differ - slower trend, z-score carry, F12 value. A one-size template would have understated Aluminium; per-metal selection is necessary.
- **Discipline matters.** removing the carry look-ahead, charging costs on the real traded price, and demanding walk-forward survival are what separate the real edges (trend, carry-momentum, z-score carry, V1 value) from the traps (level-carry over-statement, 1-day carry change, V2 reversal).
- **Value is a portfolio component, not a stand-alone.** value is weak alone, especially over the trailing five years, but earns its place through negative correlation to momentum. The portfolio, not the sleeve, is the product.

Caveats

- the per-metal best momentum pair and carry/value winners were chosen in-sample; their out-of-sample numbers hold up, but selection itself was not walk-forward.
- Value's full-sample edge is concentrated in the 2020-2021 dislocation.
- results are pre-financing, single-contract, ± 1 unit sizing; no volatility targeting at the sleeve level.
- all figures reproduce live on the dashboard; the metal toggle on each of the four signal tabs switches the entire tab to that metal's configuration.

All performance figures verified against the live dashboard engine (PnL on F1_continuous, costs on F1_raw, active-day Sharpe, no look-ahead). Copper sample ends 2025-12-31; Aluminium sample ends 2026-05-19.